SH1011 Modern fysik, HT 2022

Föreläsning 9

- Repetition
- Spin
- Mångelektronatomer.
- Uteslutningsprincipen.
- Periodiska systemet.
- Energinivåer vid screening.
- Röntgenspektrum, emission och absorption.
- Moseleys lag.
- Hur fungerar röntgenavbildning som mammografi och datortomografi.
- (Hur fungerar avbildning med magnetresonans (MR)) flyttas till kärnfysik

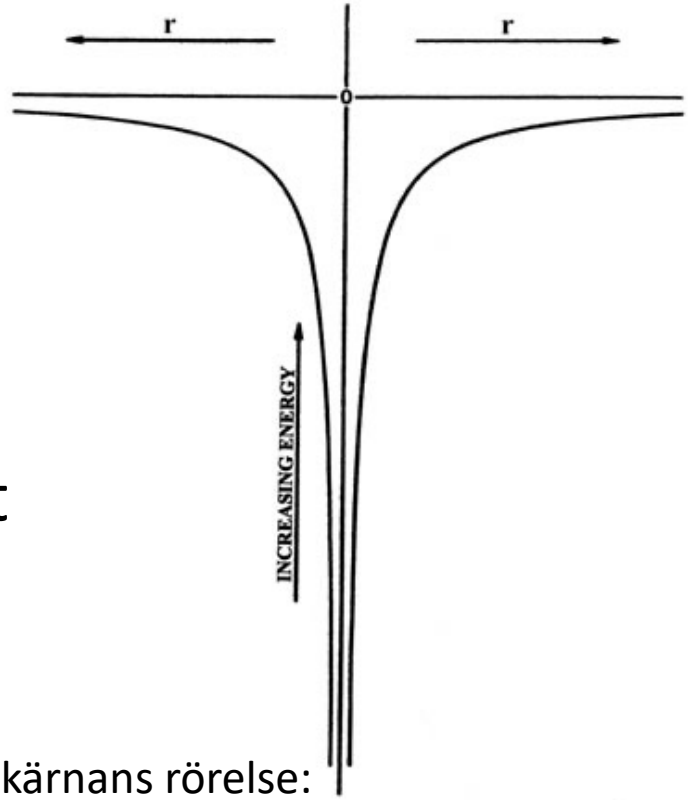


Geek3 https://en.wikipedia.org/wiki/Hydrogen_atom#/media/File:Hydrogen_eigenstate_n4_l3_m1.png

Väteatomen: kvanttal och energinivåer

- $\psi(\mathbf{r}) = \psi_{n,l,m_l}(\mathbf{r})$

- $n = 1,2,3, \ldots$ huvudkvanttalet
- $l = 0,1,2, \ldots, n-1$ bankvanttalet
- $m_l = -l, \ldots, -1,0,1, \ldots, l$ magnetiska kvanttalet



$$E_n = -\frac{me^4}{2(4\pi\epsilon_0)^2\hbar^2}\frac{1}{n^2} = \frac{-13{,}6\ \text{eV}}{n^2}$$

Korrektion för kärnans rörelse:
Byt ut m mot reducerade massan

$$m_r = \frac{m_1 m_2}{m_1 + m_2}$$

Väteatomen: stationära tillstånd

$$\psi_{2,0,0} = \frac{1}{4\sqrt{2\pi}a_0^{3/2}}\left(2 - \frac{r}{a_0}\right)e^{-r/2a_0}, \qquad \text{(2s)}$$

$$\psi_{1,0,0}(r) = \frac{1}{\sqrt{\pi}a_0^{3/2}}e^{-r/a_0} \qquad \text{(1s)}$$

$$\psi_{2,1,0} = \frac{1}{4\sqrt{2\pi}a_0^{3/2}}\frac{r}{a_0}e^{-r/2a_0}\cos\theta,$$

$$\psi_{2,1,\pm1} = \mp\frac{1}{8\sqrt{\pi}a_0^{3/2}}\frac{r}{a_0}e^{-r/2a_0}\sin\theta\, e^{\pm i\varphi} \qquad \text{(2p)}$$

$a_0 = \frac{4\pi\epsilon_0\hbar^2}{me^2}$ = 0.0529 nm

Bohr-radien

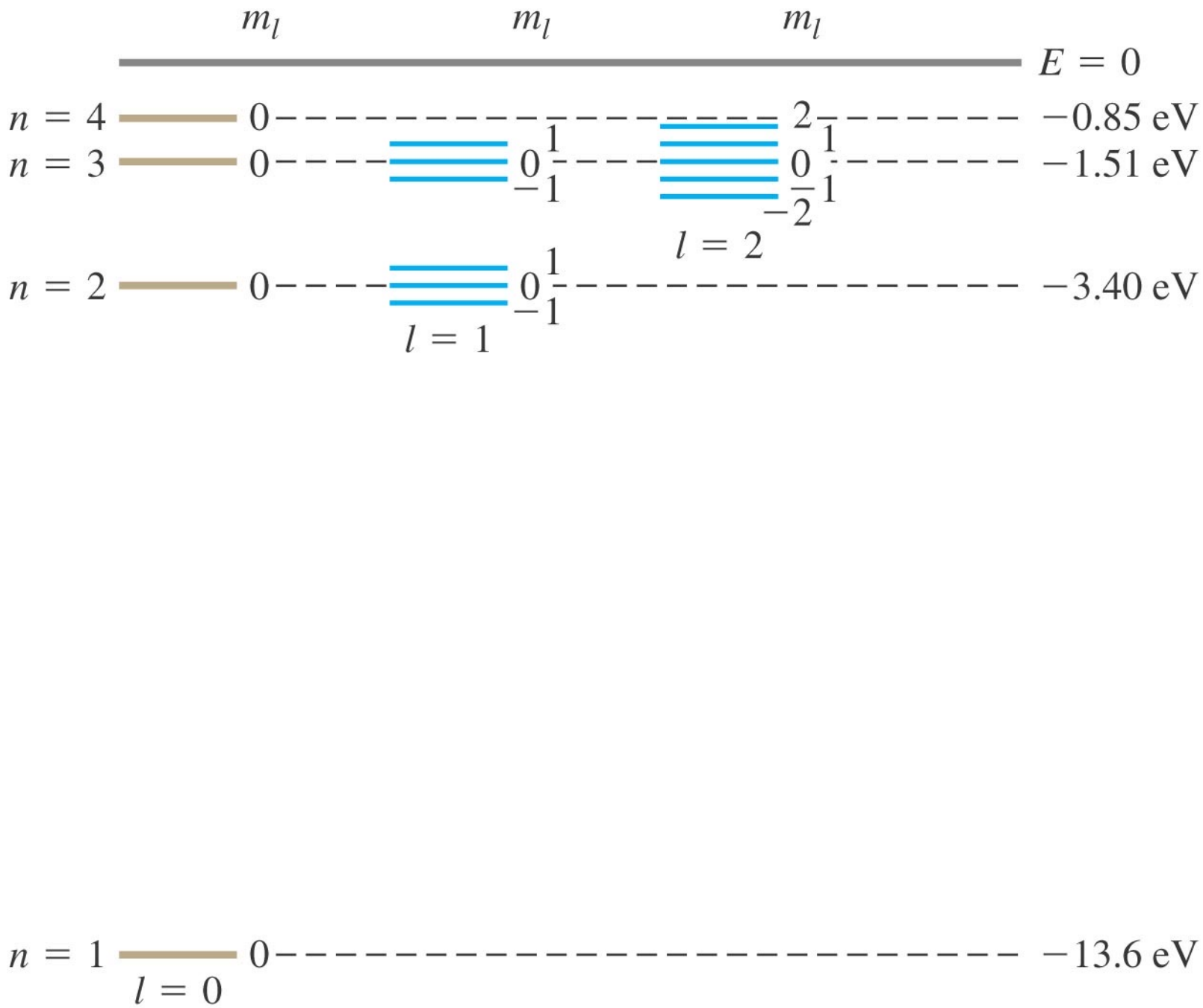
Och så vidare…

https://en.wikipedia.org/wiki/Hydrogen_atom

Väteliknande joner

He^{+}, Li^{2+}, Be^{3+} etc.

$$E_n = -Z^2 \frac{13{,}6\ \text{eV}}{n^2}$$



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Väteatomens orbitaler

$l = 0$	s	“sharp”
$l = 1$	p	“principal”
$l = 2$	d	“diffuse”
$l = 3$	f	“fundamental”
$l \geq 4$	g,h,i,k,l…	



1s



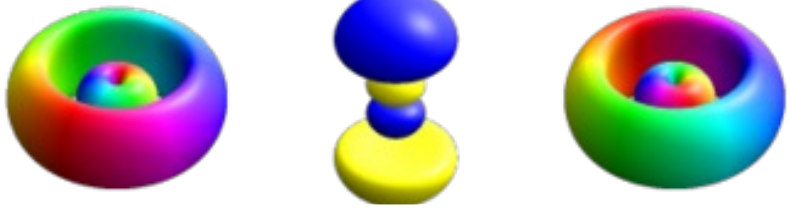
2s



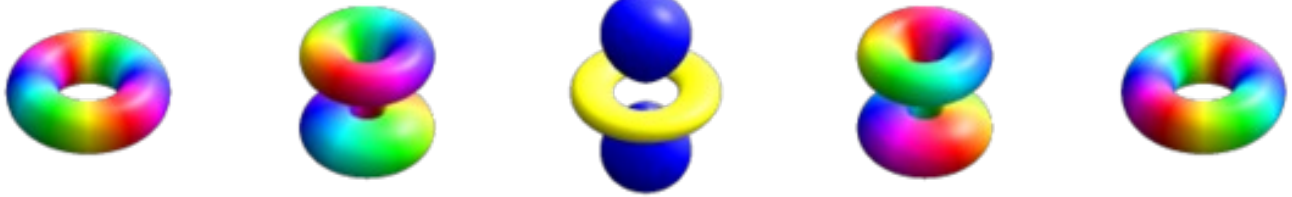
2p



3s



3p



3d

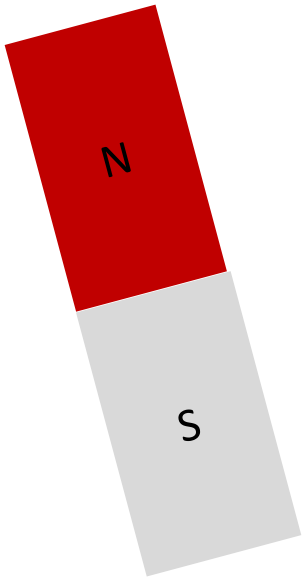
Geek3 https://en.wikipedia.org/wiki/Hydrogen-like_atom#/media/File:Atomic_orbitals_n1234_m-eigenstates.png

Magnetisk dipol

$\mu = IA$



=

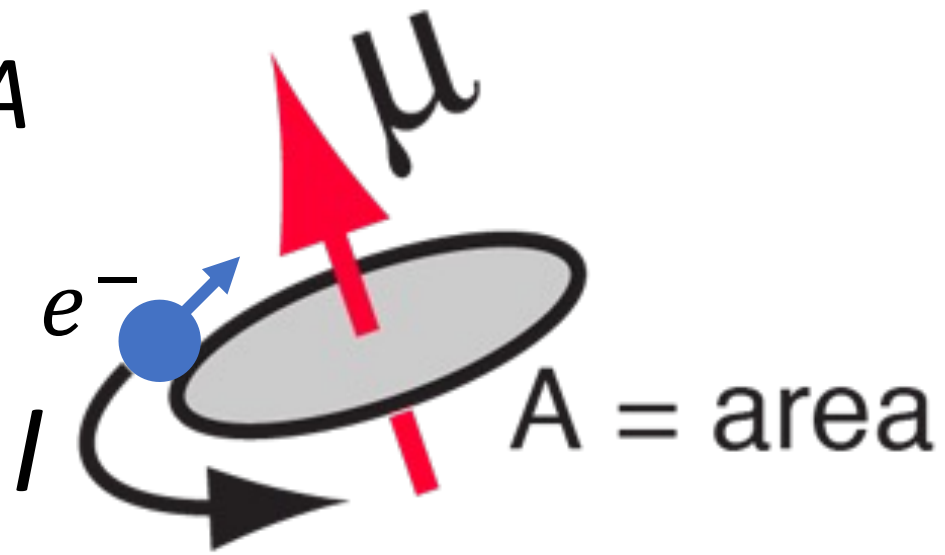


Gyromagnetisk kvot

Förhållandet mellan magnetiskt moment och rörelsemängdsmoment

$$\gamma = \mu/L$$

$\mu = IA$



I har motsatt riktning mot elektronens rörelse!

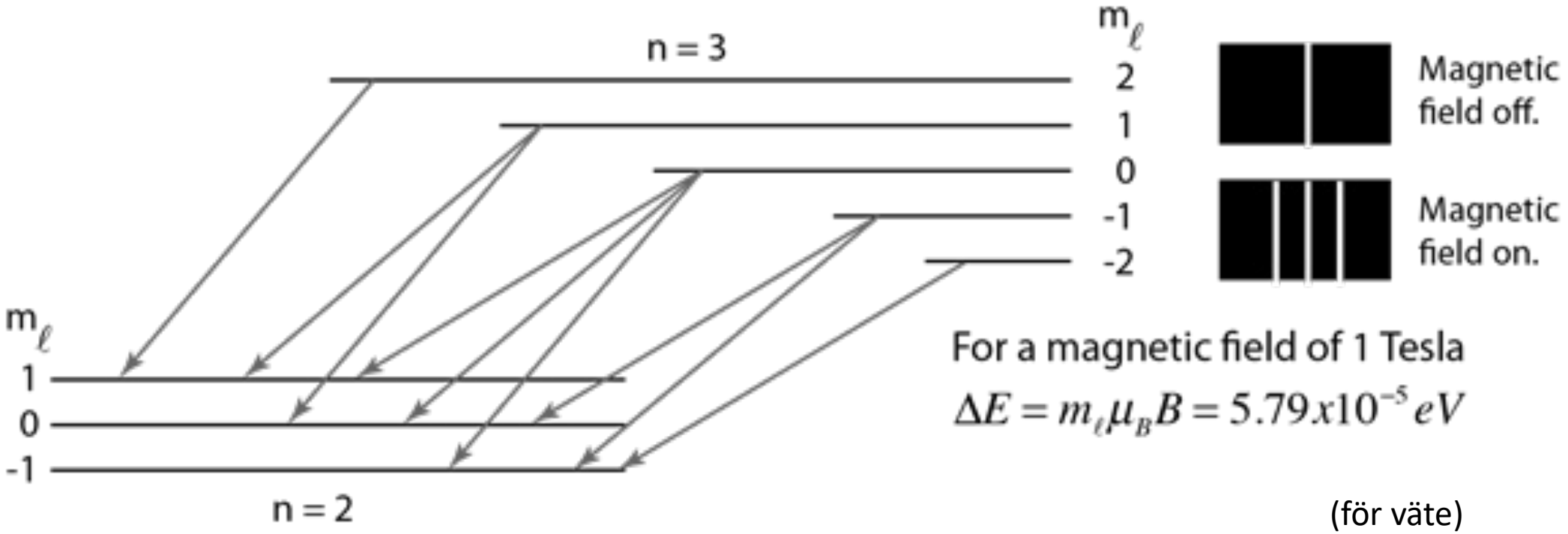
- Rörelsemängdsmomentets magnitud

$$L = \sqrt{l(l+1)}\hbar$$

- Rörelsemängdsmomentets komponent i z-led:

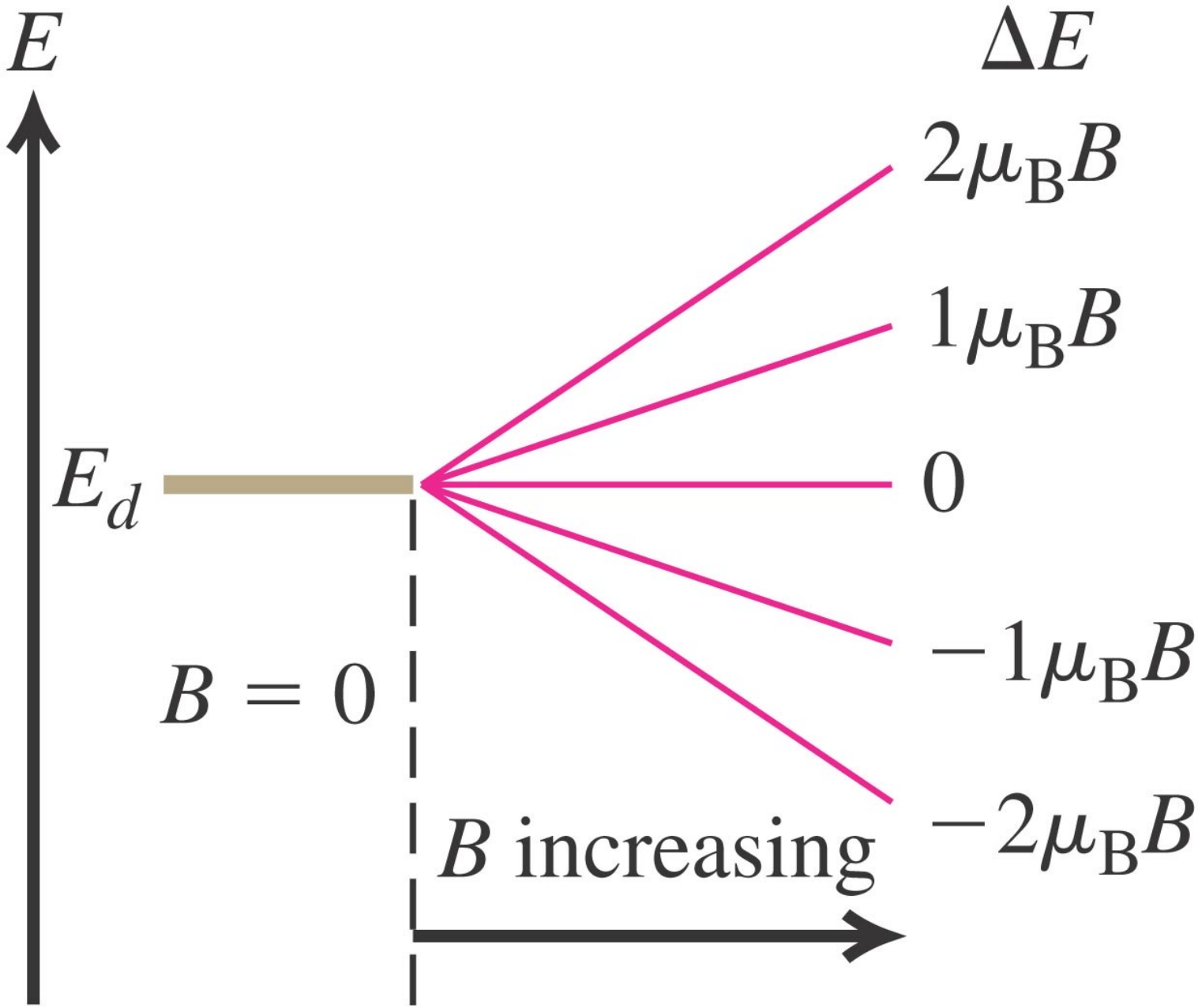
$$L_z = m_l \hbar$$

Zeemaneffekten



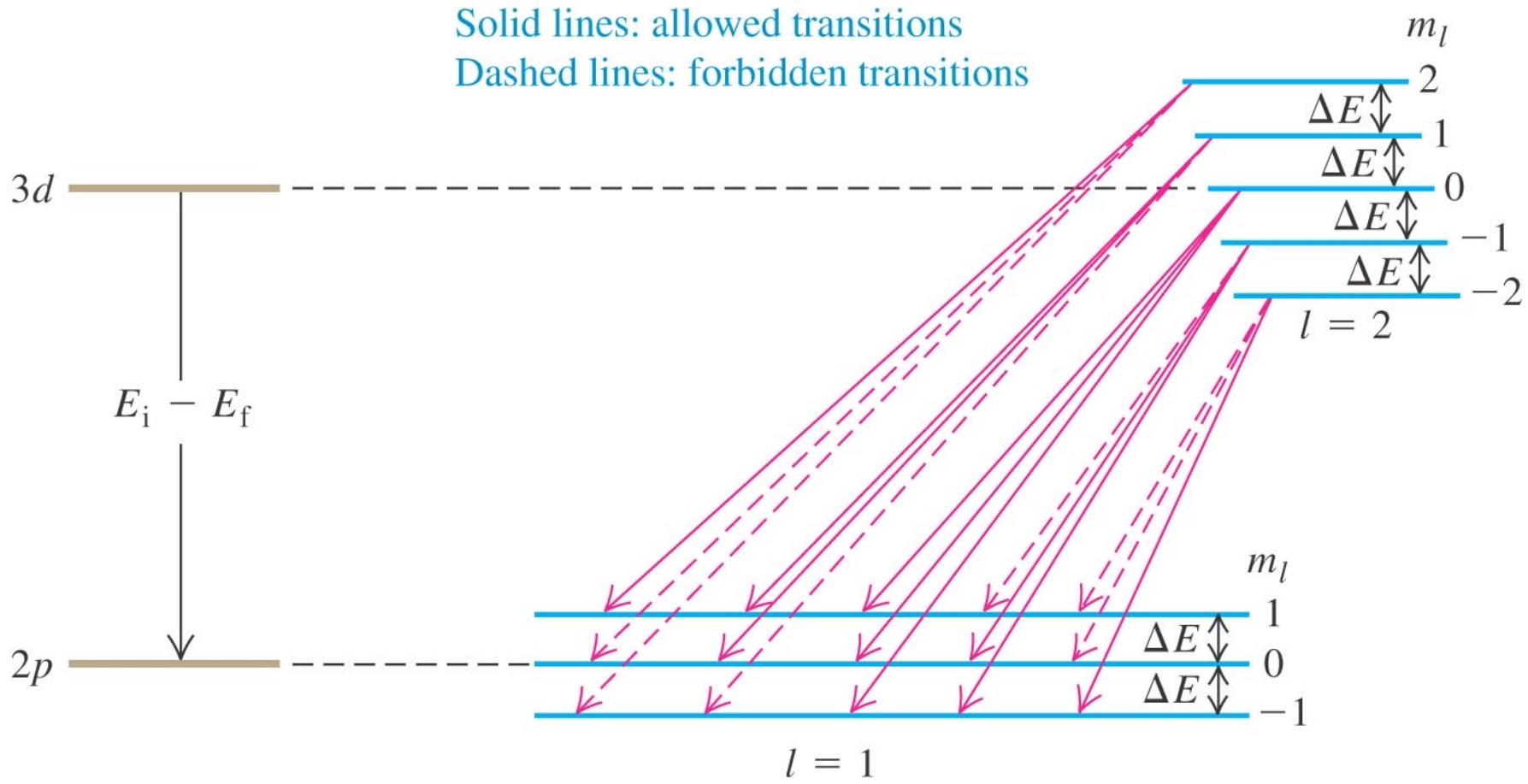
(för väte)

http://hyperphysics.phy-astr.gsu.edu/hbase/quantum/zeeman.html



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Exempel på elektronövergångar mellan olika energinivåer



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Spin (för elektron)

Spin – ett kvanttal till

$$S = \sqrt{\frac{1}{2}\left(\frac{1}{2}+1\right)}\,\hbar$$

$$S_z = m_s \cdot \hbar \qquad m_s = \pm\frac{1}{2}$$

$$\mu_z = -2.00232\,\frac{e}{2m}\,S_z$$

Spin – ban koppling

elektron som rör sig kring atomkärnan skapar ett magnetfält som μ_z växelverkar med

$$U = \mu_z \cdot B = 2.00232\,\frac{e\hbar}{2m}\,m_s = \pm\frac{1}{2}$$

Spin (för elektron)

$$S = \sqrt{\frac{1}{2}\left(\frac{1}{2} + 1\right)}\hbar$$

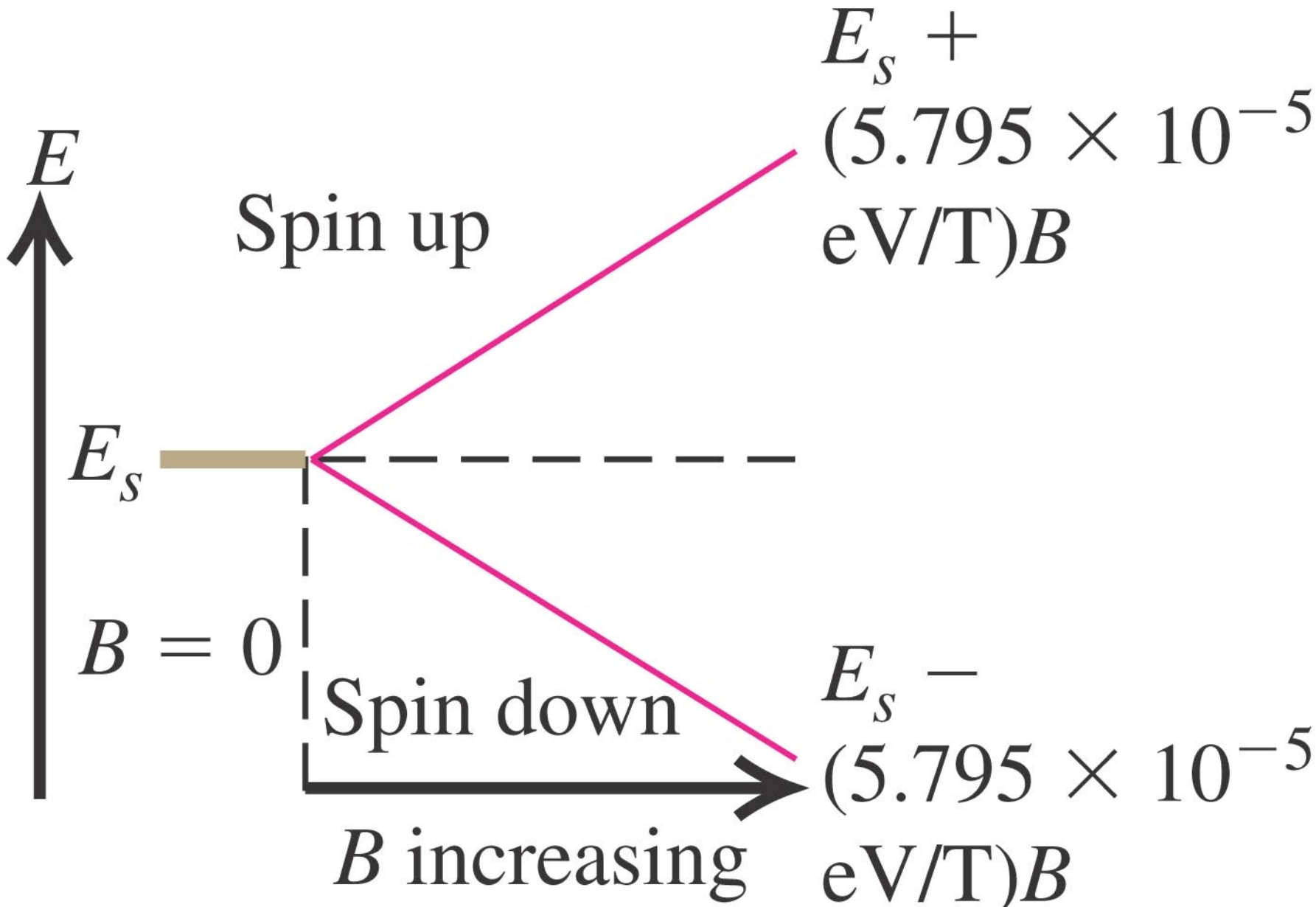
$$S_z = \pm\frac{1}{2}\hbar$$

Magnetisk dipol (elektriskt laddad partikel med rörelsemängdsmoment) växelverkar med externt magnetfält

$$U = -\mu_z B = m_l \frac{e\hbar}{2m} B$$

$$(m_l = 0, \pm 1, \pm 2, \ldots, \pm l)$$

För spin gäller samma samband men multiplicerat med g-faktor 2
Samt att m_l ersätts med spin kvanttalet +1/2, -1/2



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Tillåtna kvanttal för väteatomen
Paulis uteslutningsprincip

$$n \geq 1$$

$$0 \leq l \leq n - 1$$

$$|m_l| \leq l$$

$$m_s = \pm \frac{1}{2}$$

Tillstånd i väteatomen

Table 41.2 Quantum States of Electrons in the First Four Shells

n	l	m_l	Spectroscopic Notation	Number of States		Shell
1	0	0	$1s$	2		K
2	0	0	$2s$	2	8	L
2	1	$-1, 0, 1$	$2p$	6		
3	0	0	$3s$	2		
3	1	$-1, 0, 1$	$3p$	6	18	M
3	2	$-2, -1, 0, 1, 2$	$3d$	10		
4	0	0	$4s$	2		
4	1	$-1, 0, 1$	$4p$	6		
4	2	$-2, -1, 0, 1, 2$	$4d$	10	32	N
4	3	$-3, -2, -1, 0, 1, 2, 3$	$4f$	14		

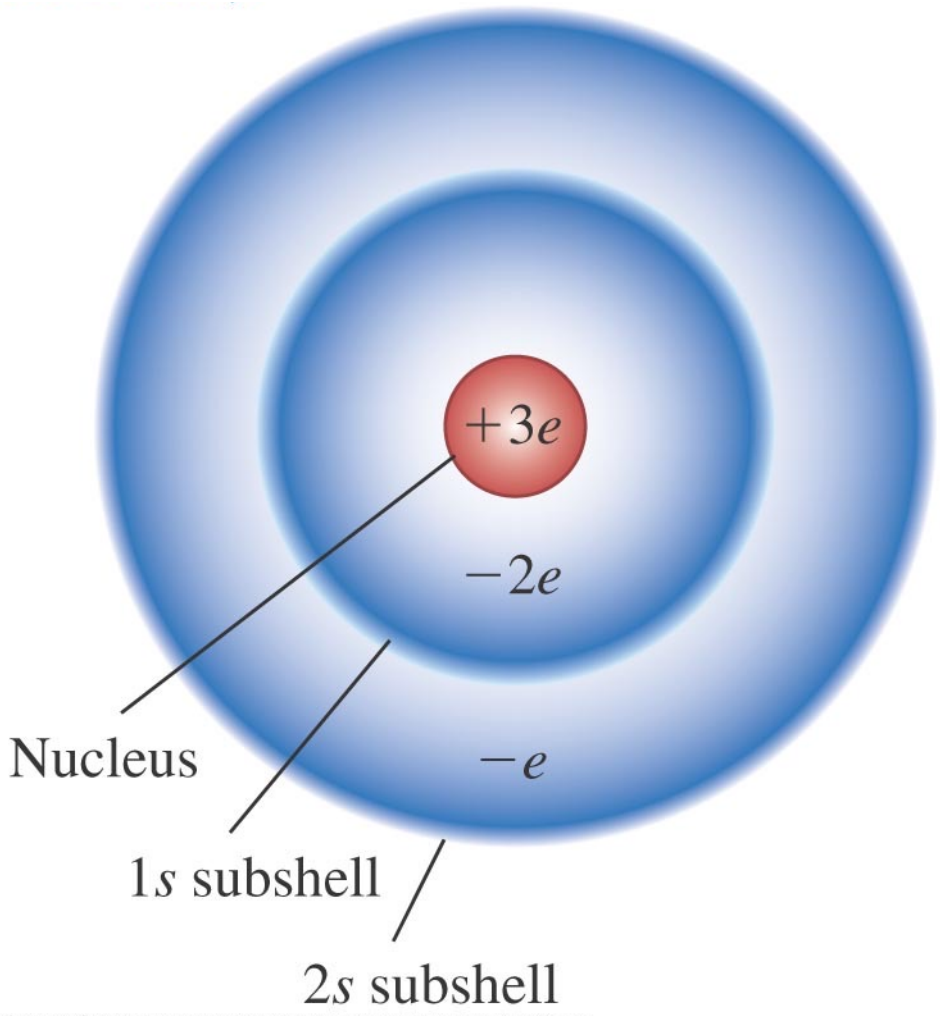
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Table 41.3 Ground-State Electron Configurations

Element	Symbol	Atomic Number (Z)	Electron Configuration
Hydrogen	H	1	$1s$
Helium	He	2	$1s^2$
Lithium	Li	3	$1s^2 2s$
Beryllium	Be	4	$1s^2 2s^2$
Boron	B	5	$1s^2 2s^2 2p$
Carbon	C	6	$1s^2 2s^2 2p^2$
Nitrogen	N	7	$1s^2 2s^2 2p^3$
Oxygen	O	8	$1s^2 2s^2 2p^4$
Fluorine	F	9	$1s^2 2s^2 2p^5$
Neon	Ne	10	$1s^2 2s^2 2p^6$
Sodium	Na	11	$1s^2 2s^2 2p^6 3s$
Magnesium	Mg	12	$1s^2 2s^2 2p^6 3s^2$
Aluminum	Al	13	$1s^2 2s^2 2p^6 3s^2 3p$
Silicon	Si	14	$1s^2 2s^2 2p^6 3s^2 3p^2$
Phosphorus	P	15	$1s^2 2s^2 2p^6 3s^2 3p^3$
Sulfur	S	16	$1s^2 2s^2 2p^6 3s^2 3p^4$
Chlorine	Cl	17	$1s^2 2s^2 2p^6 3s^2 3p^5$
Argon	Ar	18	$1s^2 2s^2 2p^6 3s^2 3p^6$
Potassium	K	19	$1s^2 2s^2 2p^6 3s^2 3p^6 4s$
Calcium	Ca	20	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$
Scandium	Sc	21	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d$
Titanium	Ti	22	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^2$
Vanadium	V	23	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^3$
Chromium	Cr	24	$1s^2 2s^2 2p^6 3s^2 3p^6 4s 3d^5$
Manganese	Mn	25	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^5$
Iron	Fe	26	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$
Cobalt	Co	27	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^7$
Nickel	Ni	28	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^8$
Copper	Cu	29	$1s^2 2s^2 2p^6 3s^2 3p^6 4s 3d^{10}$
Zinc	Zn	30	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10}$

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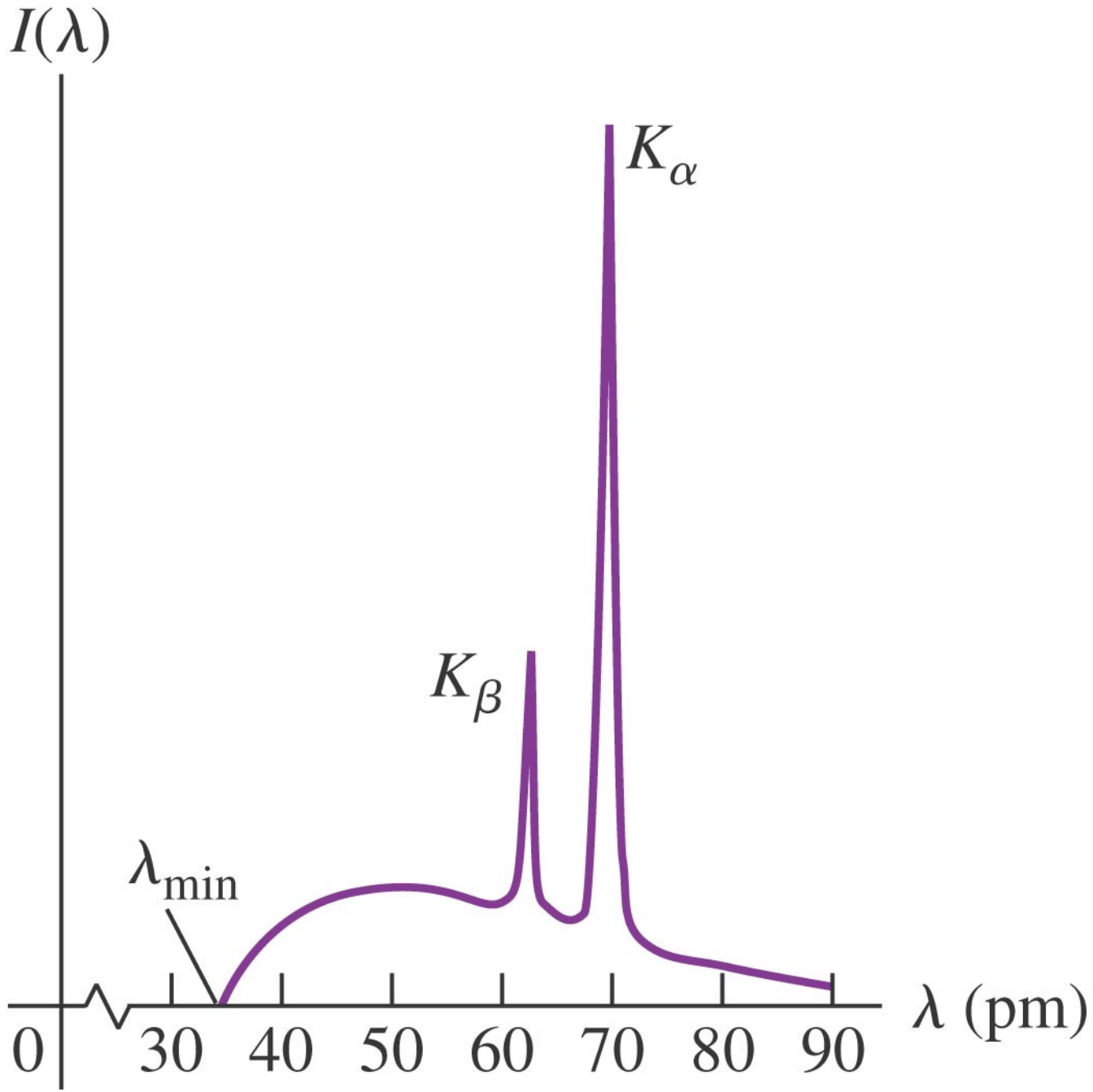
Screening



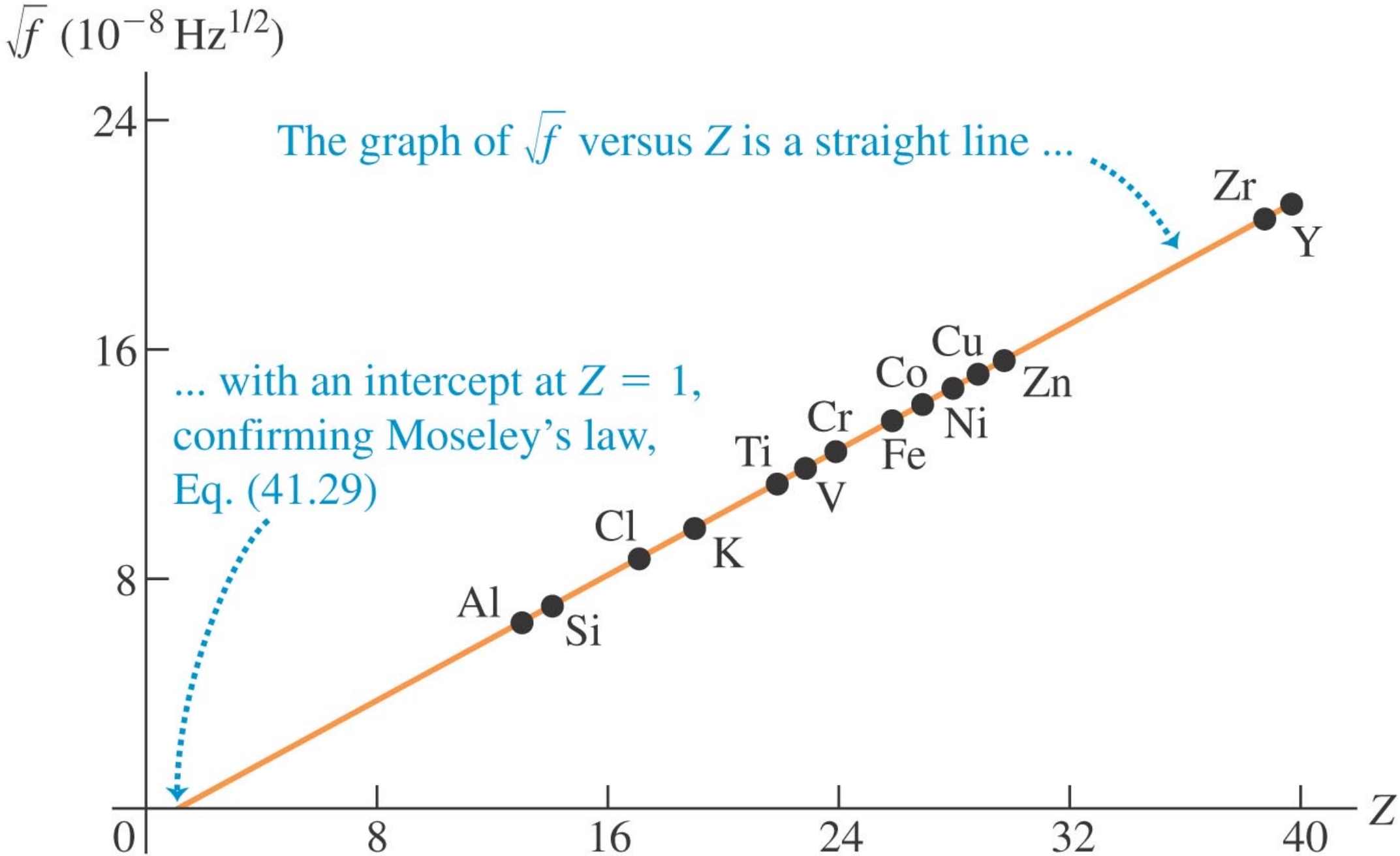
Energinivåer vid skärmning (screening)

Z_{eff}=Z-1 för en elektron i övre energitillstånd när en K-skals elektron saknas

$$E_n = -\frac{Z_{\text{eff}}^2}{n^2}(13.6\ \text{eV})$$



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Moseley's lag

$$f = (2.48 \times 10^{15}\ \text{Hz})(Z - 1)^2$$

$$E_n = -\frac{Z_{\text{eff}}^2}{n^2}(13.6\ \text{eV})$$

Moseley's lag

$$f = (2.48 \times 10^{15}\ \text{Hz})(Z-1)^2$$

$$E_n = -\frac{Z_{\text{eff}}^2}{n^2}(13.6\ \text{eV})$$

$E = hf \Rightarrow$

$f = \frac{E}{h}$

For K_α

$$E_i - E_f = -\frac{(Z-1)^2}{2^2}(13.6\text{eV}) - \left(-\frac{(Z-1)^2}{1^2}(13.6\text{eV})\right) =$$

$$= (Z-1^2) \cdot \frac{3}{4} \cdot 13.6\ \text{eV} = (Z-1)^2 \frac{3}{4} \cdot \frac{13.6 \cdot 1.602 \cdot 10^{-19}}{6.625 \cdot 10^{-34}}$$

$$= (2.46 \cdot 10^{15}\ \text{Hz})(Z-1)^2$$

Henry Moseley



the British government instituted a policy of not allowing its prominent and promising scientists to enlist for combat duty in the armed forces of the Crown…

Röntgenabsorption

